Enhancing Wind Power Forecasting Accuracy in Canada Using a Solar Data-Enhanced Hybrid Machine Learning Model: Integrating ANN, LSTM, and SVR

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Abstract*— As the world moves toward sustainable energy solutions, wind power emerges as a pivotal renewable energy (RE) source due to its accessibility and zero carbon emission. However, its unpredictable nature poses significant forecasting challenges, that impact energy management efficiency. This study tackles this vital challenge by integrating solar power data into advanced machine learning models to enhance the forecast accuracy and quantifying uncertainty of wind power.** ***For this work, the MERRA-2 dataset – a comprehensive atmospheric reanalysis from NASA – spanning 2017 to 2019 across three different locations of Canadian provinces, British Colombia, Manitoba, and Nova Scotia has been considered. A novel hybrid machine learning framework that combines the strengths of Artificial Neural Networks (ANN), Long Short-Term Memory Networks (LSTM), and Support Vector Machines (SVM), has been announced. Utilizing this framework excels in pattern recognition, temporal data processing, and regression analysis, effectively will improve the precision of wind power forecasts. Besides, it provides a robust framework for quantifying forecast uncertainty, thereby enhancing decision-making in renewable energy management. The superiority of this model is demonstrated through comparative evaluations against conventional methods using various metrics to establish its efficacy and applicability in real-world scenarios.

Keywords—Renewable Energy, Wind, Solar, Machine Learning, MERRA 2

I. INTRODUCTION

Renewable energy sources have increasingly become vital to global strategies in the quest for sustainable energy solutions. Among all of them, wind power stands out due to its considerable potential and environmental benefits. In recent years, different countries have ramped up their investment in wind power for the mitigation of impacts of climate change, and carbon footprints, and reduced their reliability on fossil fuels[1]. The International Energy Agency (IEA) highlights that rapid technological advancements have not only reduced the costs of renewable energy production but have also made wind and solar energy increasingly competitive against conventional fossil fuels. Despite all of the improvements, one of the main challenges in harnessing wind power which is its inherent intermittency still remains. Features like wind speed and direction have an enormous impact on the energy production. The unprecedented nature of wind features emphasizes the necessity of using advanced forecasting and management strategies to address grid integration issues, investment risks, and operational efficiency.

Recent advancements in forecasting methodologies have elevated the capabilities of ML algorithms to address the unpredictability of wind generation. For instance, this study [2] explores the use of ANN to predict wind speed and direction, two important factors in wind generation the wind speed was considered in this work, demonstrating a significant improvement over traditional ones. Furthermore, the integration of the LSTM network, highlighted by this work [3] has offered promising results in capturing the temporal dependencies in wind speed to refine the precision of wind power prediction. The incorporation of SVM, into wind power forecasting has also been studied in this paper [4], when combined with other ML techniques, it can effectively model the non-linearity between the wind speed and the power. The role of environmental data, like solar irradiance, air temperature, and precipitation, in enhancing the accuracy of wind power forecasts has been increasingly acknowledged. A thorough study [5] demonstrated how incorporating solar power data into a wind forecasting model could signify the precision in the accuracy of wind power by solar interconnection. Alongside that, this paper [6] also proposed different hybrid combinations of Adaptive Neuro – Fuzzy Interference Systems (ANFIS), Recurrent Kalman Filter (RKF), and Neuro-Wavelet (WNN), to demonstrate the superior performance of hybrid models over the single method ones.

The organization of this work is as follows. Section II introduces each ML technique and its integration. Section III is about data integration analysis and forecasting models. In section IV, model implementation and validation have been discussed, Results and future works are in sections V and VI, respectively.

II. INTEGRATED DATA ANALYSIS AND FORECASTING MODELS

A. Data Collection and Preparation

The approach begins with the in-depth collection of environmental data from different geographical regions within Canada, British Colombia, Manitoba, and Nova Scotia. In the below table, the locations in those provinces have been indicated.

Table 1 Data locations

Location	Latitude(Lat)	Longitude(Lon)
British Colombia	55.0013	-125.0024
Manitoba	55.0013	-97.0010
Nova Scotia	45.1960	-63.1654

This study evaluates three distinct wind turbines, each selected for their exceptional power generation capabilities, pioneering designs, and technological advancements that align with the objectives of enhancing wind power forecasting[7].

- **Enercon E126/6500**: This turbine features an impressive 126-meter rotor diameter and a gearless direct-driven system, capable of generation of 7 MW of power. It is most specifically designed for high-wind area environments, making it an ideal candidate for regions in Canada that experience intense wind conditions.
- **Siemens SWT-3.6:** As another beneficial turbine for windy areas, this turbine with a 120-meter rotor diameter and a medium capacity of 3.6 MW, has been chosen.
- **Vestas V90-2:** With a 90-meter rotor diameter and a 2 MW capacity, this turbine is well-suited for areas with moderate wind speeds, and a complementary part for this study's need to test under diverse environmental conditions.

In this study, the Enercon E126/6500, Siemens SWT-3.6, and Vetas-2 wind turbines have been used in three distinct locations in Canada, each of them carefully selected in order to showcase a broad spectrum of technologies and capabilities. This strategic deployment across diverse Canadian landscapes allows for an exhaustive analysis of turbine performance under various wind conditions (from high wind areas to variable ones).

Table 2 Turbine models and their characteristics

Turbine Model	Hub height	Rotor Diameter	Capacity
Enercon E126	136 m	126 m	7.5 Mw
SWT-3.6	90 m	120 m	3.6 Mw
V90	80 m	90 m	2 Mw

Since 1980, the MERRA-2, developed by NASA's Global Modeling and Assimilation Office, has faced fundamental enhancements, making it the backbone of many atmospheric data analysis studies. Also, NASA's ozone profile has been used in this dataset since 2004, which enriched its superiority. The Fidelity of the dataset is sustained by improvements in both Gridpoint Statistical Interpolation (GSI) and Goddard Earth Observing System (GEOS), maintaining its resolution at 50km in the latitudinal direction[9]. The integration of this data set into this study, due to its refined spatial and temporal resolution, which contains various environmental features like air temperature or wind speed, is inevitable. Incorporating vast satellite observations with advanced data adjustments aids this dataset in gathering accurate and consistent valuable information from the atmosphere. This evolutionary enhancement in data quality enriches the understanding of climate dynamics, underpinning the robustness of this study's forecasting by providing thorough information on the features of atmospheric conditions that have an enormous impact on renewable energy generation.

B. Data Cleaning and Integration

This section will explore the precise processes of ensuring the dataset quality, consistency, and comparability. Using powerful programming languages like Python aims to refine the raw data gathered into a coherent, analysis-ready format. Some issues like missing values, outliers, and inconsistencies across the dataset were issued, ensuring a solid foundation for accurate forecasting. The data source variety leads to the fact that the ML system would face different units of measurement, normalization, and transformation to establish a robust essential for quality assurance. In the following parts, each step taken to prepare the dataset for the ML model training has been clarified.

- **Missing Data Handling:** Each specific dataset inevitably contains missing values for certain features, that can introduce inaccuracies or errors in the trained model. To address this missing values are interpolated by the usage of existing data from the dataset, ensuring a seamless flow of information for the usage in models. [11].
- **Data Normalizations and Transformations:** Normalization of data processes into a standard range ensures uniformity in data units and facilitates the comparison of various scale features. [12].
- **Integration Techniques:** Integration of heterogeneous data sources presents significant challenges, especially in aligning disparate formats and scales. In this study, an exclusive The MERRA-2 dataset, known for its extensive information, along with other environmental data from various locations for the three-year span required substantial manual adjustments to ensure the uniformity of the dataset.

C. Model Configuration

In this section, intricate elaboration will be on the selected ML models for the study. For this work, ANN, SVR, and LSTMs have been chosen. The complexity of the data-set, with the varied variables and temporal dynamics, needed a profound model can find the patterns among the features. For the ANN and LTSM models, careful consideration was to their hyperparameters, hidden layers or neurons within each layer, to capture the correlation among the input datasets, thereby enhancing predictive performance. The number of hidden layers in the ANN model was designed in a way to commence with a size roughly half of the input features, to employ a geometric progression for decreasing the numbers of subsequent layers. Using this way aids in gradually concentrating the most useful information for the task.

Activation functions are essential mathematical tools in machine learning, particularly within neural networks. These functions are applied to the input of each node and transfer it into an output signal that serves as the input for the next layer. Commonly used activation functions include Rectified Linear Units (ReLU), sigmoid, and SoftMax, each of them offering unique features that would be used in different types of modeling scenarios[14].

$$f(x) = max\ (0, x) \tag{1}$$

$$f(x) = \frac{1}{1 + e^{-x}} \tag{2}$$

For the equation (1) is defined as the input and the output would be x itself, if the input is bigger than zero, and it would be zero if the input is negative. For equation (2), the function would have the input and map the output value between 0 and 1, making it suitable for better interpretation, especially for small or big numbers.

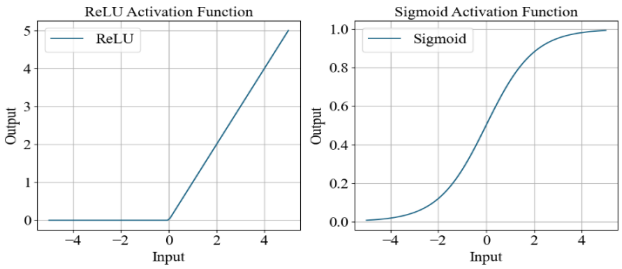


Fig. 2. ReLU and Sigmoid activation functions

D. *Evaluation Metrics*

Wind power forecasting involves continuous values, that needs to be assessed with regression analysis and appropriate evaluation metrics. This study utilizes two key metrics; R-squared for its ability to explain variance in the target variable and Mean Squared Logarithmic Error (MSLE) for handling skewed distributions and balanced error evaluation. Moreover, K-fold cross validation is used to ensure the model robustness and reliability for various datasets[15]

MSLE measure the ratio between actual (y_i) and predicted ($\bar{y}_i$) values after applying the neutral logarithm, ensuring all inputs are positive by adding small constant.

$$MSLE = \frac{1}{n}\sum_{i=1}^{n}(\log(1+y_i) - \log(1+\bar{y}_i))^2 \quad (3)$$

R2 quantifies the proportion of variance in the mean ($\bar{y}$) and other related variables, as mentioned in the last formula, in the following one.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \quad (4)$$

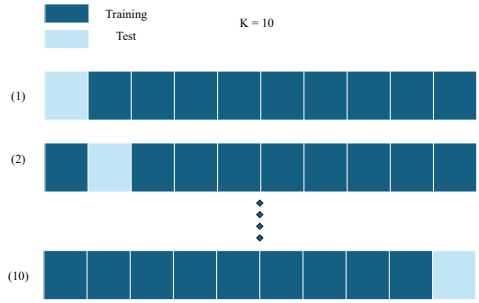


Fig. 3. K (10) - fold cross validation

K-fold cross validation evaluates the skill of ML models by dividing the dataset into K subsets. The performance measure (P) for each fold, where P_i represents the performance metrics of R2 and MSLE in each subset.

$$P = \frac{1}{K}\sum_{i=1}^{K} P_i \quad (5)$$

III. Integrating ANN, SVR, and LTSM for Predictive Modeling in Wind Power Forecasting

Forecasting has faced an enormous advance with the advent of ML. This transformation mainly relies on the vast amount of data that could be gathered by industries that are collected in datasets or websites like “Renewable Ninja”, which researchers and engineers have access to[16]. Among all ML techniques, this work focused on three specific ones. ANN, SVR (the regression model of SVM), and LSTM were chosen for their ability to handle the non-linear and intricate patterns characteristic of wind data[2]. Having these three models as a hybrid ML framework offers a promising approach for optimal RE forecasting.

- **ANN:** Utilizing the pattern recognition capabilities inherent in ANN, inspired by the neural architectures of animal brains, this model excels in identifying intricate patterns and dependencies within complex, dynamic datasets, such as those encountered in wind power time series[17].
- **SVR:** Chosen for its proficiency in managing high-dimensional data, SVR applies the kernel trick to effectively tackle the non-linear relationships found in wind power data, optimizing the balance between model complexity and predictive accuracy essential in volatile environments [5].
- **LSTM:** As a specialized form of Recurrent Neural Networks (RNNs), LSTM is particularly adept at capturing long-term dependencies in time series data, a critical feature in the predictability of wind patterns [18].

IV. Model Implementation and Validation

A. *Data Preparation and Preprocessing*

The foundation of this model implementation began with the data collection phase, where MERRA-2 was used , which is acclaimed for its comprehensive atmospheric reanalysis capabilities. The spatial resolution of the MERRA-2 data is approximately 0.5° x 0.625° (latitude x longitude), which equates to about 50km x 50km at the equator, providing detailed geographical coverage sufficient to capture diverse climatic conditions across Manitoba, Nova Scotia, and British Columbia spinning the years 2017 to 2019. One hour time interval for 1095 days from the MERRA-2 as inputs for network training, 10 × 236314 numbers of dataset with their within features, and wind generation taken as the target data. They have divided into 70% for training, 15% for validation, and the rest for testing. The data characterized included metrics from three different turbines: Enercon E126 6500, Siemens SWT 3.6, and V90 2000. The specifics of steps and input/output data can be seen in Fig. 4 and Fig. 5.

B. *Model Architecture and Hybrid Model Formulation*

The work offered the integration of two sophisticated neural networks: the ANN and LSTM. The ANN component was designed with an input layer of 256 neurons, followed by two hidden layers each comprising 128 neurons, all in ReLU activation function, and culminating in a linear activation function as an output layer. The aim was this structure could capture the complex relationship among the features. Parallelly, the LSTM model has 128 neurons as its first layer, followed by 64 neurons at another one, all in ReLU, and lastly finished by one linear activation function hidden layer. The usage of this model helped to modify the deep temporal dependencies among features. A novel aspect of this work was the integration of the last two mentioned models, serving as a

composite input for the SVR. The training part of input data has been injected into both LSTM and ANN.

C. Model Training and Validation

In this comprehensive ML approach, raw data first undergoes standardization using StandardScaler and it would be reshaped into three-dimensional arrays to meet the LSTM model requirements, alongside the ANN, both would be trained on this preprocessed data, reserving 20% as a validation set to guard against overfitting. Post-training, predictions from both previous models are concatenated, forming a proper input for the SVR, which acts as a meta-model. In the end, using this sophisticated, multi-staged training process will be evaluated in the next part to assess its value.

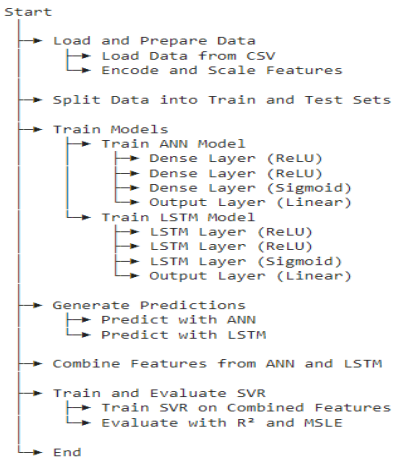


Fig. 4. Model training steps

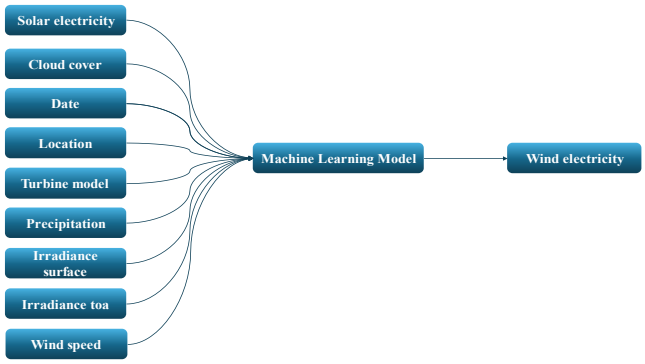


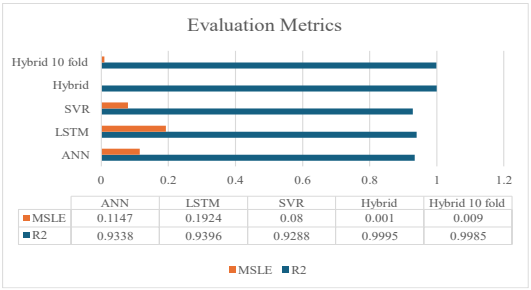
Fig. 5. ML inputs and outputs

D. Performance Evaluation

For the evaluation process R2 and MSLE metrics within 10-fold cross-validation framework have been used to ensure the model effectiveness, stability and reliability. Sensitivity test, involving changes in input tolerances and evaluation metrics, assesses the forecasting robustness in this work. For having various sensitivities, different 'epsilon' values in the SVR controls the epsilon-intensive tube, that impacts support vector count and regression smoothness[19].

V. Results

The results obtained from this model are presented in this section. Through a complete assessment, as can be seen in Fig. 6, different aspects of this model have been observed. The proposed machine learning framework has been shown to significantly enhance the accuracy of wind power forecasts, surpassing the traditional statistical methods. The R2 value reaches **0.9985** in this model, which is approximately higher than the most competitive one among the three mentioned methods. Moreover, for the MSLE section, the Hybrid model received **0.001**, which is considered the best against the other used model in this work.



	ANN	LSTM	SVR	Hybrid	Hybrid 10 fold
MSLE	0.1147	0.1924	0.08	0.001	0.009
R2	0.9338	0.9396	0.9288	0.9995	0.9985

Fig. 6. Evaluation metrics for the model

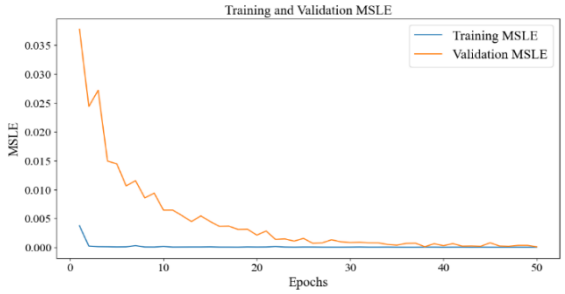


Fig. 7. Training and validation MSLE for the Hybrid

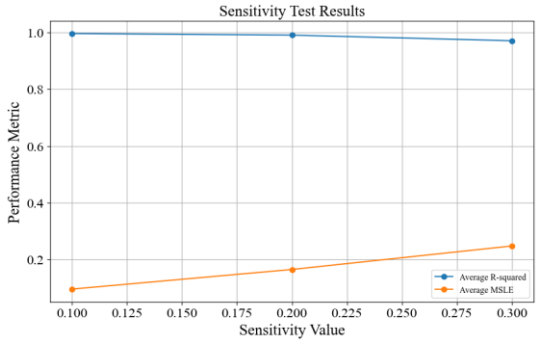


Fig. 8. Sensitivity test result for the Hybrid

For the performance, the MSLE served as a vital metric, depicting the discrepancies in predicted value versus the actual one during both training and validation. These processes have been done through 50 epochs, that have been plotted in Fig. 7, revealing the constant reduction in error, thus understanding the model's ability to generalize and learn the data effectively.

For the sensitivity check of inputs, an analysis has been performed to measure the model's responsiveness to variations in inputs. The visualization of three different SVR's epsilon parameters, 0.1, 0.2, and 0.3, has been demonstrated

in Fig. 8, which shows the robustness and relative influence on the predictive outcome. The necessity of this experiment was to identify how varying the sensitivity, which here is the SVR, could impact the prediction. Maintaining high R^2 and small MSLE, yet slightly increasing during the sensitivity test, impacts the successful capturing of underlying data dynamics and model adoption at managing proportional divergences.

Lastly in Fig. 9, a comparison between the real value and the prediction ones after using the hybrid model has been illustrated representing the power of this model in using this specific dataset over one day in the Alberta for Enercon E126/6500 turbine.

The architecture of the proposed model, combining the strength of ANN, LSTM, and SVR, leads to more reliable and precise forecasting. The results underscore the model's robustness throughout this process. R^2 and MSLE have been used in this work, alongside the implementation of K-fold cross-validation for validating the proposed work. The sensitivity analysis, revealed the model's adaptability to varying input parameters, particularly within SVR, by considering different 'epsilons'. Using this method, evaluate the model flexibility and responsiveness against the input variations.

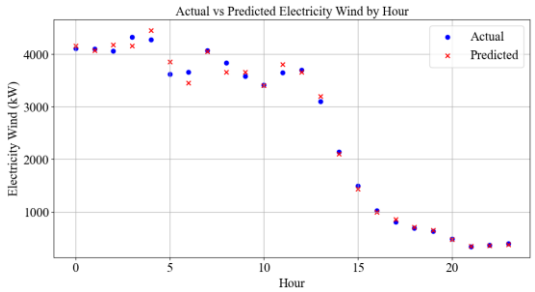


Fig. 9. Final prediction vs actual values for the Hybrid (UTC time)

VI. Future works

The proposed Hybrid model, in the field of R2 and MSLE, achieved better results than other methods. This improvement suggests that this model could provide more reliable predictions, potentially reducing the operational risks and cost in the wind power management. However, it is important to note that, the current model has been tested under conditions specific to the Canadian provinces of British Colombia, Manitoba, and Nova Scotia. Based on the success of the hybrid model, future works could benefit from exploring several avenues for enhancement and refinement. One potential area of focus could be the integration of additional environmental factors, such as atmospheric pressure, to expose its beneficial impact on the model performance and accuracy. Moreover, considering different locations or various turbines could provide valuable insights into the generalizability and applicability of the hybrid model in diverse situations. Last but not least, combining other ML methods, exclusively deep learning ones, could aid the model in obtaining a better understanding of the dataset for further accurate and wholesome predictions.

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